RESEARCH DRAFT

Contents

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1. Loops and suspension

Let $\mathcal{C}$ be a pointed ∞-category, with zero object $*$, and suppose that it has the finite limits and colimits required below.

1A. **Fibers and cofibers.** For $f\colon X \to Y$, its fiber over the canonical basepoint $0\colon * \to Y$ is defined by the cartesian square

[] %%| filename: fiber-square %%| additionalPackages: \usepackage{amsmath,amssymb,tikz-cd} \begin{tikzcd} \operatorname{Fib}(f) \arrow[r,"p_X"] \arrow[d,"p_*"'] \arrow[dr,phantom,very near start,"\lrcorner"] & X \arrow[d,"f"]\\ * \arrow[r,"0"'] & Y \end{tikzcd}

The fiber-product notation for its apex is

$$\operatorname{Fib}(f) = X \times_Y *.$$

The cofiber is defined by the cocartesian square

[] %%| filename: cofiber-square %%| additionalPackages: \usepackage{amsmath,amssymb,tikz-cd} \begin{tikzcd} X \arrow[r,"f"] \arrow[d,"0"'] \arrow[dr,phantom,very near end,"\ulcorner"] & Y \arrow[d,"i_Y"]\\ * \arrow[r,"i_*"'] & \operatorname{Cof}(f) \end{tikzcd}

The pushout notation for its apex is

$$\operatorname{Cof}(f) = Y \amalg_X *.$$

These definitions specialize in pointed spaces to the usual homotopy fiber and homotopy cofiber.

1B. **Loops and suspension.** For an object X of $\mathcal{C}$, loops are defined by the pullback square

[] %%| filename: loop-square %%| additionalPackages: \usepackage{amsmath,amssymb,tikz-cd} \begin{tikzcd} \Omega X \arrow[r,"\operatorname{pr}_2"] \arrow[d,"\operatorname{pr}_1"'] \arrow[dr,phantom,very near start,"\lrcorner"] & * \arrow[d,"0"]\\ * \arrow[r,"0"'] & X \end{tikzcd}

Suspension is defined by the pushout square

[] %%| filename: suspension-square %%| additionalPackages: \usepackage{amsmath,amssymb,tikz-cd} \begin{tikzcd} X \arrow[r,"0"] \arrow[d,"0"'] \arrow[dr,phantom,very near end,"\ulcorner"] & * \arrow[d,"j_1"]\\ * \arrow[r,"j_2"'] & \Sigma X \end{tikzcd}

Thus

$$\Omega X = * \times_X *, \qquad \Sigma X = * \amalg_X *.$$

In pointed spaces these are the usual loop-space and reduced-suspension constructions.

1C. **Arrow categories and path spaces.** For a higher category C, evaluation at the endpoints of the constructed walking arrow gives

$$[[1], C] \longrightarrow C \times C.$$

The fiber over (x, y) is the hom-category $[x, y]_C$; applying Π_∞ gives $\operatorname{Map}_C(x, y)$. The fiber over (x, x) is the endomorphism category of x, and its full subcategory on the invertible objects is the automorphism category of x.

1D. **The suspension-loop adjunction.** When $\mathcal{C}$ has finite limits and colimits, suspension is left adjoint to loops:

$$\operatorname{adj} \mathcal{C}\mathcal{C}\Sigma\Omega, \qquad \Sigma \dashv \Omega.$$

For $\mathcal{C} = \mathcal{S}_*$, the ∞-category of pointed spaces, this recovers the classical adjunction

$$\operatorname{Map}_*(\Sigma X, Y) \simeq \operatorname{Map}_*(X, \Omega Y).$$

1E. **Fiber sequences.** A composable pair $F \to E \to B$ is a fiber sequence when F is equivalent to the homotopy fiber over a specified basepoint of B. Applying homotopy groups gives the long exact sequence. Its component-level portion is the pointed-set sequence recorded in **sec-pi0-fiber**.